

HIRAPINE TABLET 200MG

DESCRIPTION:

A round 9mm, scored tablet white to off-white in colour with markings 'DUO 3' on the scored side and 'DUO 4' on the other side.

COMPOSITION:

Each tablet contains Nevirapine 200mg.

PHARMACODYNAMICS:

Nevirapine is a non-nucleoside reverse transcriptase inhibitor (NNRTI) of HIV-1. Nevirapine binds directly to reverse transcriptase (RT) and blocks the RNA-dependent and DNA-dependent DNA polymerase activities by causing a disruption of the enzyme's catalytic site. The activity of nevirapine does not compete with template or nucleoside triphosphates. HIV-2 RT and eukaryotic DNA polymerases (such as human DNA polymerases α , β , γ or δ) are not inhibited by nevirapine.

PHARMACOKINETICS:

Nevirapine is readily absorbed following oral administration and absorption is not affected by food. Peak plasma concentration occurs up to 4 hours after a single dose. Nevirapine is about 60% bound to plasma proteins. Concentrations in the CNS are about 45% of those in plasma. Nevirapine crosses the placenta and is distributed into breast milk. It is extensively metabolised by hepatic microsomal enzymes, principally by cytochrome P450 isoenzymes of the CYP3A family. Autoinduction of these enzymes results in a 1.5- to 2-fold increase in apparent oral clearance after 2 to 4 weeks administration of usual doses, and a decrease in terminal half-life from 45 hours to 25 to 30 hours over the same period. Nevirapine is mainly excreted in the urine as glucuronide conjugates of the hydroxylated metabolites.

INDICATION:

Nevirapine in combination with antiretroviral agents is indicated for the treatment of HIV-1 infection in adults and children over the age of 2 months. Resistant virus emerges rapidly when Nevirapine is administered as monotherapy or in dual combination therapy with an antiretroviral agent. Therefore, Nevirapine should always be administered in combination with at least two additional antiretroviral agents. Avoid usage of Nevirapine in patient with CD4+cell count greater than 250 cells/mm³.

RECOMMENDED DOSAGE:

Adults 16 years and older: The recommended dose is nevirapine 200 mg daily for the first 14 days (this lead-in period should be used because it has been found to lessen the frequency of rash), followed by 200 mg twice daily, in combination with nucleoside analogue antiretroviral agents. For concomitantly administered nucleoside therapy, the manufacturer's recommended dosage and monitoring should be followed.

Children aged 8 years up to 16 years: The recommended dose is nevirapine 4 mg/kg once daily for two weeks followed by 4 mg/kg twice daily thereafter. The total daily dose should not exceed 400 mg of nevirapine.

Children aged 2 months up to 8 years: The recommended dose is nevirapine 4 mg/kg once daily for two weeks followed by 7mg/kg twice daily thereafter. The total daily dose should not exceed 400 mg of nevirapine.

Dosage Management Considerations: Patients should be advised of the need to take Nevirapine every day as prescribed. If a dose is missed the patient should not double the next dose but should take the next dose as soon as possible.

Clinical chemistry tests, which include liver function tests, should be performed prior to initiating Nevirapine therapy and at appropriate intervals during therapy.

Nevirapine administration should be discontinued if patients experience severe rash or a rash accompanied by constitutional symptoms. Patients experiencing rash during the 14-day lead-in period of 200 mg/day (4 mg/kg/day for children) should not have their Nevirapine dose increased until the rash has resolved (see Precautions, Information for Patients).

Nevirapine administration should be interrupted in patients experiencing moderate or severe liver function test abnormalities (excluding GGT) until liver function tests have returned to baseline. Nevirapine may then be restarted at 200 mg daily increasing to 200 mg twice daily with caution, after extended observation. Nevirapine should be permanently discontinued if moderate or severe liver function test abnormalities recur. If clinical hepatitis occurs, characterised by anorexia, vomiting, icterus AND laboratory findings (such as moderate or severe liver function test abnormalities (excluding GGT)), Nevirapine must be permanently stopped. Nevirapine should not be readministered to patients who have required permanent discontinuation for clinical hepatitis due to Nevirapine.

Patients who interrupt Nevirapine dosing for more than 7 days should restart the recommended dosing, using 200 mg daily (4 mg/kg/day for children) for the first 14 days (lead-in) followed by 200 mg twice daily (4 or 7 mg/kg twice daily for children depending on age).

No data are available to recommend a dosage of Nevirapine in patients with hepatic dysfunction, renal insufficiency, or undergoing dialysis.

ROUTE OF ADMINISTRATION: Oral

CONTRACINDICATIONS:

Hypersensitivity to the active substance or to any of the excipients.

NEVIRAPINE should not be readministered to patients who have required permanent discontinuation for severe rash, rash accompanied by constitutional symptoms, hypersensitivity reactions, or clinical hepatitis due to nevirapine.

NEVIRAPINE should not be used in patients with severe hepatic impairment or pre-treatment AST or ALT > 5 ULN until baseline AST/ALT are stabilised < 5 ULN.

NEVIRAPINE should not be readministered in patients who previously had AST or ALT > 5 ULN during NEVIRAPINE therapy and had recurrence of liver function abnormalities upon readministration of NEVIRAPINE.

Herbal preparations containing St John's Wort (*Hypericum perforatum*) must not be used while taking NEVIRAPINE due to the risk of decreased plasma concentrations and reduced clinical effects of nevirapine.

The available pharmacokinetic data suggest that the concomitant use of rifampicin and NEVIRAPINE is not recommended.

WARNINGS & PRECAUTIONS:

On the basis of pharmacodynamic data NEVIRAPINE should only be used with at least two other antiretroviral agents.

The first 18 weeks of therapy with NEVIRAPINE are a critical period which requires close monitoring of patients to disclose the potential appearance of severe and life-threatening skin reactions (including cases of Stevens-Johnson syndrome and toxic epidermal necrolysis) or serious hepatitis/hepatic failure. The greatest risk of hepatic events and skin reactions occurs in the first 6 weeks of therapy. However, the risk of any hepatic event continues past this period and monitoring should continue at frequent intervals. Female gender and higher CD4 counts at the initiation of therapy place patients at greater risk of hepatic adverse events. Unless the benefit outweighs the risk NEVIRAPINE should not be initiated in adult females with CD4 cell counts greater than 250 cells/mm³ or in adult males with CD4 cell counts greater than 400 cells/mm³. This is based on the occurrence of serious and life threatening hepatotoxicity in controlled and uncontrolled studies.

In some cases, hepatic injury has progressed despite discontinuation of treatment. Patients developing signs or symptoms of hepatitis, severe skin reaction or hypersensitivity reactions must discontinue NEVIRAPINE and seek medical evaluation immediately. Nevirapine should not be restarted following severe hepatic, skin or hypersensitivity reactions.

The dosage must be strictly adhered to, especially the 14-days lead-in period.

Cutaneous reactions

Severe and life-threatening skin reactions, including fatal cases, have occurred in patients treated with NEVIRAPINE mainly during the first 6 weeks of therapy. These have included cases of Stevens-Johnson syndrome, toxic epidermal necrolysis and hypersensitivity reactions characterised by rash, constitutional findings and visceral involvement. Patients should be intensively monitored during the first 18 weeks of treatment. Patients should be closely monitored if an isolated rash occurs. NEVIRAPINE must be permanently discontinued in any patient experiencing severe rash or a rash accompanied by constitutional symptoms (such as fever, blistering, oral lesions, conjunctivitis, facial oedema, muscle or joint aches, or general malaise), including Stevens-Johnson syndrome, or toxic epidermal necrolysis. NEVIRAPINE must be permanently discontinued in any patient experiencing hypersensitivity reaction (characterised by rash with constitutional symptoms, plus visceral involvement, such as hepatitis, eosinophilia, granulocytopenia, and renal dysfunction).

NEVIRAPINE administration above the recommended dose might increase the frequency and seriousness of skin reactions, such as Stevens-Johnson syndrome and toxic epidermal necrolysis.

Concomitant prednisone use (40 mg/day for the first 14 days of NEVIRAPINE administration) has been shown not to decrease the incidence of NEVIRAPINE-associated rash, and may be associated with an increase in incidence and severity of rash during the first 6 weeks of NEVIRAPINE therapy.

Some risk factors for developing serious cutaneous reactions have been identified, they include failure to follow the initial dosing of 200 mg daily during the lead-in period and a long delay between the initial symptoms and medical consultation. Women appear to be at higher risk than men of developing rash, whether receiving NEVIRAPINE or non-NEVIRAPINE containing therapy.

Patients should be instructed that a major toxicity of NEVIRAPINE is rash. They should be advised to promptly notify their physician of any rash and avoid delay between the initial symptoms and medical consultation. The majority of rashes associated with NEVIRAPINE occur within the first 6 weeks of initiation of therapy. Therefore, patients should be monitored carefully for the appearance of rash during this period. Patients should be instructed that dose escalation is not to occur if any rash occurs during the two-week lead-in dosing period, until the rash resolves.

Any patient experiencing severe rash or a rash accompanied by constitutional symptoms such as fever, blistering, oral lesions, conjunctivitis, facial oedema, muscle or joint aches, or general malaise should discontinue medication and immediately seek medical evaluation. In these patients NEVIRAPINE must not be restarted.

If patients present with a suspected NEVIRAPINE-associated rash, liver function tests should be performed. Patients with moderate to severe elevations (AST or ALT > 5 ULN) should be permanently discontinued from NEVIRAPINE.

If a hypersensitivity reaction occurs, characterised by rash with constitutional symptoms such as fever, arthralgia, myalgia and lymphadenopathy, plus visceral involvement, such as hepatitis, eosinophilia, granulocytopenia, and renal dysfunction, NEVIRAPINE should be permanently stopped and not be re-introduced.

Hepatic reactions

Severe and life-threatening hepatotoxicity, including fatal fulminant hepatitis, has occurred in patients treated with NEVIRAPINE. The first 18 weeks of treatment is a critical period which requires close monitoring. The risk of hepatic events is greatest in the first 6 weeks of therapy. However the risk continues past this period and monitoring should continue at frequent intervals throughout treatment. Increased AST or ALT levels ≥ 2.5 ULN and/or co-infection with hepatitis B and/or C at the start of antiretroviral therapy is associated with greater risk of hepatic adverse reactions during antiretroviral therapy in general, including NEVIRAPINE containing regimens. Female gender and patients with higher CD4 counts are at increased risk of hepatic adverse events.

Women have a threefold higher risk than men for symptomatic, often rash-associated, hepatic events (5.8% versus 2.2%), and patients with higher CD4 counts at initiation of NEVIRAPINE therapy are at higher risk for symptomatic hepatic events with NEVIRAPINE. In a retrospective review, women with CD4 counts >250 cells/mm³ had a 12 fold higher risk of symptomatic hepatic adverse events compared to women with CD4 counts <250 cells/mm³ (11.0% versus 0.9%). An increased risk was observed in men with CD4 counts > 400 cells/mm³ (6.3% versus 1.2% for men with CD4 counts <400 cells/mm³).

Patients should be informed that hepatic reactions are a major toxicity of NEVIRAPINE requiring close monitoring during the first 18 weeks. They should be informed that occurrence of symptoms suggestive of hepatitis should lead them to discontinue NEVIRAPINE and immediately seek medical evaluation, which should include liver function tests.

Liver monitoring

Abnormal liver function tests have been reported with NEVIRAPINE, some in the first few weeks of therapy.

Asymptomatic elevations of liver enzymes are frequently described and are not necessarily a contraindication to use NEVIRAPINE. Asymptomatic GGT elevations are not a contraindication to continue therapy.

Monitoring of hepatic tests should be done every two weeks during the first 2 months of treatment, at the 3rd month and then regularly thereafter. Liver test monitoring should be performed if the patient experiences signs or symptoms suggestive of hepatitis and/or hypersensitivity.

If AST or ALT ≥ 2.5 ULN before or during treatment, then liver tests should be monitored more frequently during regular clinic visits. NEVIRAPINE should not be administered to patients with pre-treatment AST or ALT > 5 ULN until baseline AST/ALT are stabilised < 5 ULN.

Physicians and patients should be vigilant for prodromal signs or findings of hepatitis, such as anorexia, nausea, jaundice, bilirubinuria, acholic stools, hepatomegaly or liver tenderness. Patients should be instructed to seek medical attention promptly if these occur.

If AST or ALT increase to > 5 ULN during treatment, NEVIRAPINE should be immediately stopped. If AST and ALT return to baseline values and if the patient had no clinical signs or symptoms of hepatitis, rash, constitutional symptoms or other findings suggestive of organ dysfunction, it may be possible to reintroduce NEVIRAPINE, on a case by case basis, at the starting dosage regimen of 200 mg/day for 14 days followed by 400 mg/day. In these cases, more frequent liver monitoring is required. If liver function abnormalities recur, NEVIRAPINE should be permanently discontinued. If clinical hepatitis occurs, characterised by anorexia, nausea, vomiting, icterus AND laboratory findings (such as moderate or severe liver function test abnormalities (excluding GGT), NEVIRAPINE must be permanently stopped. NEVIRAPINE should not be readministered to patients who have required permanent discontinuation for clinical hepatitis due to nevirapine.

Liver disease

The safety and efficacy of NEVIRAPINE has not been established in patients with significant underlying liver disorders. NEVIRAPINE is contraindicated in patients with severe hepatic impairment. Patients with chronic hepatitis B or C and treated with combination antiretroviral therapy are at an increased risk for severe and potentially fatal hepatic adverse events. In the case of concomitant antiviral therapy for hepatitis B or C, please refer also to the relevant product information for these medicinal products.

Patients with pre-existing liver dysfunction including chronic active hepatitis have an increased frequency of liver function abnormalities during combination antiretroviral therapy and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered.

Post-exposure-prophylaxis

Serious hepatotoxicity, including liver failure requiring transplantation, has been reported in HIV-uninfected individuals receiving multiple doses of NEVIRAPINE in the setting of post-exposure-prophylaxis (PEP), an unapproved use. The use of NEVIRAPINE has not been evaluated within a specific study on PEP, especially in term of treatment duration and therefore, is strongly discouraged.

Other warnings

Combination therapy with NEVIRAPINE is not a curative treatment of patients infected with HIV-1; patients may continue to experience illnesses associated with advanced HIV-1 infection, including opportunistic infections.

The long-term effects of nevirapine are unknown at this time. Combination therapy with NEVIRAPINE has not been shown to reduce the risk of transmission of HIV-1 to others through sexual contact or contaminated blood.

Combination antiretroviral therapy has been associated with the redistribution of body fat (lipodystrophy) in HIV infected patients. The long-term consequences of these events are currently unknown.

Knowledge about the mechanism is incomplete. A connection between visceral lipomatosis and PIs and lipatrophy and NRTIs has been hypothesised.

A higher risk of lipodystrophy has been associated with individual factors such as older age, and with drug related factors such as longer duration of antiretroviral treatment and associated metabolic disturbances. Clinical examination should include evaluation for physical signs of fat redistribution. Consideration should be given to the measurement of fasting serum lipids and blood glucose. Lipid disorders should be managed as clinically appropriate.

Nevirapine may interact with some medicinal products; therefore, patients should be advised to report to their doctor the use of any other medications.

Oral contraceptives and other hormonal methods of birth control should not be used as the sole method of contraception in women taking NEVIRAPINE, since nevirapine might lower the plasma concentrations of these medications. For this reason, and to reduce the risk of HIV transmission, barrier contraception (e.g., condoms) is recommended. Additionally, when oral contraceptives are used for hormonal regulation during administration of NEVIRAPINE the therapeutic effect should be monitored.

Pharmacokinetic results suggest caution should be exercised when NEVIRAPINE is administered to patients with moderate hepatic dysfunction and should not be administered in patients with severe hepatic dysfunction. Overall, the results suggest that patients with mild to moderate hepatic dysfunction, defined as Child-Pugh Classification Score ≤ 7 , do not require an adjustment in NEVIRAPINE dosing. In patients with renal dysfunction, who are undergoing dialysis, pharmacokinetic results suggest that supplementing NEVIRAPINE therapy with an additional 200 mg dose of NEVIRAPINE following each dialysis treatment would help offset the effects of dialysis on nevirapine clearance. Otherwise patients with $CL_{cr} \geq 20$ ml/min do not require an adjustment in NEVIRAPINE dosing.

Immune reactivation syndrome.

In HIV-infected patients with severe immune deficiency at the time of institution of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples are cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and *Pneumocystis carinii* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary.

INTERACTION WITH OTHER MEDICAMENTS:

NRTIs: No dosage adjustments are required when NEVIRAPINE is taken in combination with zidovudine, didanosine, or zalcitabine. From a reported study, when the zidovudine data were pooled from two studies ($n = 33$) in which HIV-1 infected patients received NEVIRAPINE 400 mg/day either alone or in combination with 200-300 mg/day didanosine or 0.375 to 0.75 mg/day zalcitabine on a background of zidovudine therapy, nevirapine produced a non-significant decline of 13% in zidovudine area under the curve (AUC) and a non-significant increase of 5.8% in zidovudine C_{max} . In a subset of patients ($n = 6$) who were administered NEVIRAPINE 400 mg/day and didanosine on a background of zidovudine therapy, nevirapine produced a significant decline of 32% in zidovudine AUC and a non-significant decline of 27% in zidovudine C_{max} . Paired data suggest that zidovudine had no effect on the pharmacokinetics of nevirapine. In one crossover study, nevirapine had no effect on the steady-state pharmacokinetics of either didanosine ($n = 18$) or zalcitabine ($n = 6$).

Results from a 36 day reported study in HIV infected patients ($n = 25$) administered NEVIRAPINE, nelfinavir (750 mg t.i.d.) and stavudine (30-40 mg b.i.d.) showed no statistically significant changes in the AUC or C_{max} of stavudine. Furthermore, a population pharmacokinetic study of 90 patients assigned to receive lamivudine with NEVIRAPINE or placebo revealed no changes to lamivudine apparent clearance and volume of distribution, suggesting no induction effect of nevirapine on lamivudine clearance.

Non-nucleoside reverse transcriptase inhibitors (NNRTIs): Reported study ($n=14$) showed that steady-state pharmacokinetic parameters of nevirapine were not affected by co-administration of efavirenz. However, drug levels of efavirenz were significantly reduced in the presence of nevirapine. The AUC of efavirenz decreased by 22% and the C_{min} by 36%. When co-administered with nevirapine a dose increase of efavirenz to 800mg once daily may be warranted.

PIs: Nevirapine is a mild to moderate inducer of the hepatic enzyme CYP3A; therefore, it is possible that co-administration with PIs (also metabolised by CYP3A) may result in an alteration in the plasma concentration of either agent.

Reported study ($n = 31$) of HIV infected patients administered NEVIRAPINE and saquinavir (hard gelatin capsules; 600 mg t.i.d.) indicated that their co-administration leads to a mean reduction of 24% ($p = 0.041$) in saquinavir AUC and no significant change in nevirapine plasma levels. The reduction in saquinavir levels due to this interaction may further reduce the marginal plasma levels of saquinavir which are achieved with the hard gelatin capsule formulation.

Reported study evaluated once daily dosing of saquinavir soft gel capsule (sgc) with a 100 mg dose of ritonavir. All patients concomitantly received NEVIRAPINE. The study showed that the combination of saquinavir sgc and 100 mg of ritonavir had no measurable effect on the pharmacokinetic parameters of nevirapine, compared to historical controls. The effect of nevirapine on the pharmacokinetics of saquinavir sgc in the presence of 100 mg of ritonavir, was modest and clinically insignificant.

Results from a reported study ($n = 25$) with HIV infected patients administered NEVIRAPINE and indinavir (800 mg q8h) indicated that their co-administration leads to a 28% mean decrease ($p < 0.01$) in indinavir AUC and no significant change in nevirapine plasma levels. No definitive clinical conclusions have been reached regarding the potential impact of co-administration of nevirapine and indinavir. A dose increase of indinavir to 1000 mg q8h should be considered when indinavir is given with nevirapine 200 mg b.i.d.; however, there are no data currently available to establish that the short term or long term antiviral activity of indinavir 1000 mg q8h with nevirapine 200 mg b.i.d. will differ from that of indinavir 800 mg q8h with nevirapine 200 mg b.i.d.

Results from a reported study ($n = 25$) with HIV infected patients administered NEVIRAPINE and ritonavir (600 mg b.i.d. [using a gradual dose escalation regimen]) indicated that their coadministration leads to no significant change in ritonavir or nevirapine plasma levels.

Results from a 36 day reported study in HIV infected patients ($n = 25$) administered NEVIRAPINE, nelfinavir (750 mg t.i.d.) and stavudine (30-40 mg b.i.d.) showed no statistically significant changes in nelfinavir pharmacokinetic parameters after the addition of nevirapine (AUC + 4%, C_{max} + 14% and C_{min} - 2%). Compared to historical controls nevirapine levels appeared to be unchanged.

There were no increased safety concerns noted with the coadministration of NEVIRAPINE with any of these PIs when used in combination.

There was no apparent change in the pharmacokinetics of lopinavir when used concomitantly with NEVIRAPINE in healthy volunteers. In single PI experienced patients, nevirapine, used in combination with lopinavir / ritonavir 400/100 mg (3 capsules) twice daily and NRTIs, provided very good virological response rates. Results from a pharmacokinetic study in paediatric patients revealed a decrease in lopinavir concentrations during nevirapine co-administration. The clinical significance of this interaction is unknown. However a dose increase of lopinavir / ritonavir to 533/133 mg (4 capsules or 6.5 ml) may be considered when used in combination with nevirapine in patients where reduced susceptibility to lopinavir / ritonavir is clinically suspected (by treatment history or laboratory evidence).

Ketoconazole: In one reported study, administration of nevirapine 200 mg b.i.d. with ketoconazole 400 mg q.d. resulted in a significant reduction (63% median reduction in ketoconazole AUC and a 40 % median reduction in ketoconazole C_{max}). In the same study, ketoconazole administration resulted in a 15-28% increase in the plasma levels of nevirapine compared to historical controls. Ketoconazole and NEVIRAPINE should not be given concomitantly. The effects of nevirapine on itraconazole are not known.

Fluconazole: Co-administration of fluconazole and NEVIRAPINE resulted in approximately 100% increase in nevirapine exposure compared with historical data where NEVIRAPINE was administered alone. Because of the risk of increased exposure to nevirapine, caution should be exercised if the medicinal products are given concomitantly and patients should be monitored closely. There was no clinically relevant effect of nevirapine on fluconazole.

Oral Contraceptives: As oral contraceptives should not be used as the sole method of contraception in HIV infected patients, other means of contraception (such as barrier methods) are recommended in patients being treated with NEVIRAPINE. Furthermore a pharmacokinetic interaction has been identified. Nevirapine 200 mg b.i.d. was co-administered with a single dose of an oral contraceptive containing ethinyl estradiol (EE) 0.035mg and norethindrone (NET) 1.0 mg. Compared to plasma concentrations observed prior to nevirapine administration, the median AUC for 17 α -EE was significantly decreased by 29% after 28 days of nevirapine dosing. There was a significant reduction in EE mean resident time and half-life. There was a significant reduction (18%) in median AUC for NET, without changes in mean resident time or half-life. The magnitude of the effect suggests that the dose of the oral contraceptive should be adjusted to allow adequate treatment for indications other than contraception (e.g., endometriosis), if used with nevirapine.

Other medicinal products metabolised by CYP3A: Nevirapine is an inducer of CYP3A and potentially CYP2B6, with maximal induction occurring within 2-4 weeks of initiating multiple-dose therapy. Based on the known metabolism of methadone, nevirapine may decrease plasma concentrations of methadone by increasing its hepatic metabolism. Narcotic withdrawal syndrome has been reported in patients treated with NEVIRAPINE and methadone concomitantly. Methadone-maintained patients beginning NEVIRAPINE therapy should be monitored for evidence of withdrawal and methadone dose should be adjusted accordingly.

Other compounds that are substrates of CYP3A and CYP2B6 may have decreased plasma concentrations when co-administered with NEVIRAPINE. Therefore, careful monitoring of the therapeutic effectiveness of P450 metabolised medicinal products is recommended when taken in combination with NEVIRAPINE.

CYP isoenzyme inhibitors: The results of a nevirapine-clarithromycin interaction reported study (n = 18) resulted in a significant reduction in clarithromycin AUC (30%) and C_{max} (- 21%) but a significant increase in the AUC (58%) and C_{max} (62%) of the active metabolite 14-OH clarithromycin. There was a significant increase in the nevirapine C_{min} (28%) and a non-significant increase in nevirapine AUC (26%) and C_{max} (24%). These results would suggest that no dose adjustment is necessary for either clarithromycin and NEVIRAPINE when the two medicinal products are co-administered. Close monitoring of hepatic abnormalities and activity against Mycobacterium avium-intracellular complex (MAC) is nevertheless recommended.

Monitoring of steady-state nevirapine trough plasma concentrations in patients who received long-term NEVIRAPINE treatment revealed that nevirapine trough concentrations were elevated in patients who received cimetidine (+ 7%, n = 13).

CYP isoenzyme inducers: An open-label reported study (n = 14) to determine the effects of nevirapine on the steady state pharmacokinetics of rifampicin resulted in no significant change in rifampicin C_{max} and AUC. In contrast, rifampicin produced a significant lowering of nevirapine AUC (- 58%), C_{max} (- 50 %) and C_{min} (- 68%) compared to historical data.

The available pharmacokinetic data suggest that the concomitant use of rifampicin and NEVIRAPINE is not recommended. Therefore, these medicinal products should not be used in combination. Physicians needing to treat patients co-infected with tuberculosis and using a NEVIRAPINE containing regimen may consider use of rifabutin instead. Rifabutin and NEVIRAPINE can be administered concurrently without dose adjustments (see below). Alternatively physicians may consider switching to a triple NRTI combination for a variable period of time, depending on the tuberculosis treatment regimen.

In a pharmacokinetic reported study the concomitant administration of NEVIRAPINE with rifabutin resulted in a non-significant 12% (median) increase in the steady-state AUC, a non-significant 3% decrease in C₃ and a significant 20% increase in the C_{maxss}. Non-significant changes were found on 25-O-desacetyl-rifabutin (rifabutin active metabolite) AUC, C_{minss} or C_{maxss}. A statistically significant increase in the apparent clearance of nevirapine (9%) compared to historical pharmacokinetic data was reported. This study suggests that there is no clinically relevant interaction between nevirapine and rifabutin. Therefore, the two drugs can be administered concurrently without dose adjustments provided that a careful monitoring of the adverse reactions is performed.

Warfarin: The interaction between nevirapine and the antithrombotic agent warfarin is complex, with the potential for both increases and decreases in coagulation time when used concomitantly. The net effect of the interaction may change during the first weeks of co-administration or upon discontinuation of NEVIRAPINE, and close monitoring of anticoagulation levels is therefore warranted.

Hypericum perforatum: Serum levels of nevirapine can be reduced by concomitant use of the herbal preparation St John's Wort (Hypericum perforatum). This is due to induction of drug metabolism enzymes and/or transport proteins by St John's Wort. Herbal preparations containing St John's Wort should therefore not be combined with NEVIRAPINE. If patient is already taking St John's Wort check nevirapine and if possible viral levels and stop St John's Wort. Nevirapine levels may increase on stopping St John's Wort. The dose of NEVIRAPINE may need adjusting. The inducing effect may persist for at least 2 weeks after cessation of treatment with St John's Wort.

Other information: Studies using human liver microsomes indicated that the formation of nevirapine hydroxylated metabolites was not affected by the presence of dapsone, rifabutin, rifampicin, and trimethoprim/sulfamethoxazole. Ketoconazole and erythromycin significantly inhibited the formation of nevirapine hydroxylated metabolites.

USE IN PREGNANCY & LACTATION:

There was no evidence for teratogenicity in reproductive studies performed in rats and rabbits treated with oral doses up to 50 and 300 mg/kg/day nevirapine. In rats a significant decrease in foetal body weight occurred at doses providing systemic exposure approximately 50% higher, based on AUC, than that seen at the recommended clinical dose. Maternal toxicity and observable effects on foetal development were not observed in the rat with a systemic exposure equivalent to that seen at the recommended human dose or in the rabbit with a systemic exposure approximately 50% higher than that seen at the recommended human dose. There have been no adequate and well controlled studies of nevirapine in pregnant women, nor are there reports of infants born to women who conceived while receiving nevirapine chronic dosing in clinical trials.

Nevirapine is excreted in the breast milk. See "Use in Pregnancy" for information on studies conducted on pregnant women prior to delivery. It is generally recommended that HIV-1 infected women should not breastfeed infants regardless of the use of antiretroviral agents, to avoid post-natal transmission of HIV-1.

SIDE EFFECTS:

The most frequently reported adverse events related to NEVIRAPINE therapy, across all clinical trials, were rash, nausea, fatigue, fever, headache, vomiting, diarrhoea, abdominal pain and myalgia.

The postmarketing experience has shown that the most serious adverse reactions are Stevens-Johnson syndrome and toxic epidermal necrolysis and serious hepatitis/hepatic failure and hypersensitivity reactions, characterised by rash with constitutional symptoms such as fever, arthralgia, myalgia and lymphadenopathy, plus visceral involvement, such as hepatitis, eosinophilia, granulocytopenia, and renal dysfunction. The first 18 weeks of treatment is a critical period which requires close monitoring.

The following adverse events which may be causally related to the administration of NEVIRAPINE have been reported. The frequencies estimated are based on pooled clinical trial data for events considered related to NEVIRAPINE treatment.

Frequency classes: very common (>1/10); common (>1/100, <1/10); uncommon (>1/1,000, <1/100); rare (>1/10,000, <1/1,000); very rare (<1/10,000).

Blood and lymphatic system disorders: Rare: granulocytopenia, anaemia

Immune system disorders: Common: allergic reactions. Rare: hypersensitivity (syndrome), anaphylaxis

Nervous system disorders: Common: headache

Gastrointestinal disorders: Common: nausea. Uncommon: vomiting, abdominal pain. Rare: diarrhoea

Hepato-biliary disorders: Common: hepatitis (1.2%), liver function tests abnormal. Uncommon: jaundice. Rare: liver failure / fulminant hepatitis

Skin and subcutaneous tissue disorders: Common: rash (9%). Uncommon: Stevens Johnson syndrome (0.3%), urticarial. Rare: toxic epidermal necrolysis, angio-oedema

Musculoskeletal, connective tissue and bone disorders: Uncommon: myalgia. Rare: arthralgia

General disorders and administration site conditions: Uncommon: fatigue, fever

Combination antiretroviral therapy has been associated with redistribution of body fat (lipodystrophy) in HIV infected patients including the loss of peripheral and facial subcutaneous fat, increased intraabdominal and visceral fat, breast hypertrophy and dorsocervical fat accumulation (buffalo hump).

Combination antiretroviral therapy has been associated with metabolic abnormalities such as hypertriglyceridaemia, hypercholesterolaemia, insulin resistance, hyperglycaemia and hyperlactataemia.

The following events have also been reported when NEVIRAPINE has been used in combination with other anti-retroviral agents: pancreatitis, peripheral neuropathy and thrombocytopenia. These events are commonly associated with other antiretroviral agents and may be expected to occur when NEVIRAPINE is used in combination with other agents; however it is unlikely that these events are due to NEVIRAPINE treatment. Hepatic-renal failure syndromes have been rarely reported.

In HIV-infected patients with severe immune deficiency at the time of initiation of combination antiretroviral therapy (CART), an inflammatory reaction to asymptomatic or residual opportunistic infections may arise.

Skin and subcutaneous tissues: The most common clinical toxicity of NEVIRAPINE is rash, with NEVIRAPINE attributable rash occurring in 9% of patients in combination regimens in controlled studies. In these clinical trials 24% of patients treated with a NEVIRAPINE containing regimen experienced rash compared with 15% of patients treated in control groups. Severe rash occurred in 1.7% of NEVIRAPINE-treated patients compared with 0.2% of patients treated in the control groups.

Rashes are usually mild to moderate, maculopapular erythematous cutaneous eruptions, with or without pruritus, located on the trunk, face and extremities. Allergic reactions (anaphylaxis, angioedema and urticaria) have been reported. Rashes occur alone or in the context of hypersensitivity reactions, characterised by rash with constitutional symptoms such as fever, arthralgia, myalgia and lymphadenopathy, plus visceral involvement, such as hepatitis, eosinophilia, granulocytopenia, and renal dysfunction.

Severe and life-threatening skin reactions have occurred in patients treated with NEVIRAPINE, including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN). Fatal cases of SJS, TEN and hypersensitivity reactions have been reported. The majority of severe rashes occurred within the first 6 weeks of treatment and some required hospitalisation, with one patient requiring surgical intervention.

Hepato-biliary: The most frequently observed laboratory test abnormalities are elevations in liver function tests (LFTs), including ALT, AST, GGT, total bilirubin and alkaline phosphatase. Asymptomatic elevations of GGT levels are the most frequent. Cases of jaundice have been reported. Cases of hepatitis (severe and life-threatening hepatotoxicity, including fatal fulminant hepatitis) have been reported in patients treated with NEVIRAPINE. In a large clinical trial, the risk of a serious hepatic event among 1121 patients receiving NEVIRAPINE for a median duration of greater than one year was 1.2% (versus 0.6% in placebo group). The best predictor of a serious hepatic event was elevated baseline liver function tests. The first 18 weeks of treatment is a critical period which requires close monitoring.

Paediatric patients: Based on experience of 361 paediatric patients treated in clinical trials, the most frequently reported adverse events related to NEVIRAPINE were similar to those observed in adults, with the exception of granulocytopenia which was more commonly observed in children. In post-marketing surveillance anaemia has been more commonly observed in children. Isolated cases of Stevens-Johnson syndrome or Stevens-Johnson/toxic epidermal necrolysis transition syndrome have been reported in this population.

SYMPTOMS AND TREATMENT OF OVERDOSE:

There is no known antidote for Nevirapine overdosage. Cases of Nevirapine overdose at doses ranging from 800 to 1800 mg per day for up to 15 days have been reported. Patients have experienced events including oedema, erythema nodosum, fatigue, fever, headache, insomnia, nausea, pulmonary infiltrates, rash, vertigo, vomiting and weight decrease. All events subsided following discontinuation of Nevirapine.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

SHELF LIFE:

Please refer to outer package.

PACK SIZE:

Pack in blister pack of 6 x 10's tablets in a box.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD

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